

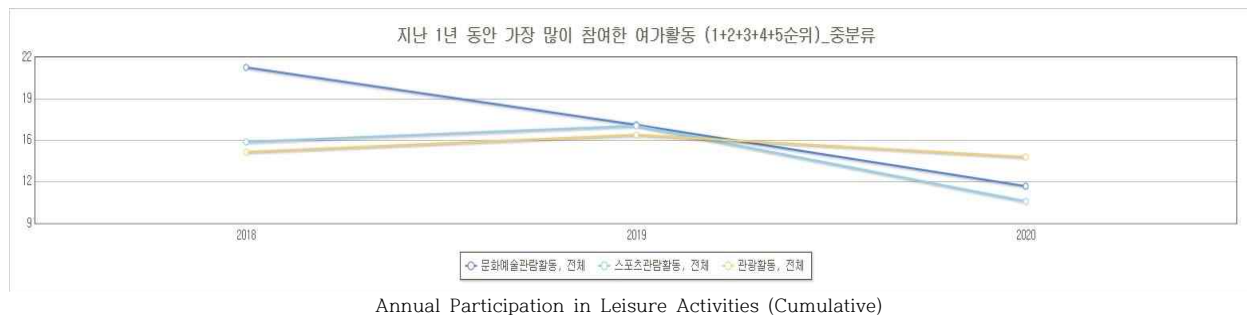
Analysis of Factors Influencing Box Office Success



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I Introduction

COVID-19 was an unprecedented event that had a profound impact across multiple sectors. In 2020, major countries such as the United States and the United Kingdom recorded fiscal deficits exceeding 10% of GDP, and the global economic downturn led to decreased growth (South Korea's GDP growth rate in Q2 2020 was -3.3%). Beyond the economy, COVID-19 brought significant changes to everyday life. International travel became difficult, and cultural activities became limited. Leisure-related industries—such as culture and arts, tourism, and sports—experienced severe economic setbacks, and the film industry was hit particularly hard. Furthermore, COVID-19 not only shrank the movie industry but also triggered structural changes across the broader content production ecosystem. In this study, I aimed to examine the existing structure of the film industry, assess whether meaningful structural changes occurred during the pandemic, and identify the key factors that drive box office success.



<Source: Statistics Korea / Original Data: Ministry of Culture, Sports and Tourism - Cultural Policy Division>

※ Prior Research

Early studies on box office performance include Litman's regression analysis using variables such as production budget, number of theaters, ratings, and distributor influence. The study found that production budget and theater count significantly affected box office results. Later, Litman expanded the explanatory variables by adding genre, while Wyatt included release timing to improve explanatory power. More recent studies incorporate online word-of-mouth indicators to explain box office success and even construct predictive models using such information.

* Characteristics of Movie Data

¹⁾Movies exhibit the characteristics of experience goods: consumers cannot fully evaluate the quality before watching, leading to heavy reliance on external information. This increases dependence on word-of-mouth and online reviews. Movies require long production periods and follow an investment-production-distribution structure. Box office success is typically evaluated using revenue or audience count.

* Structural Changes in the Film Industry

With people spending more time at home during the pandemic, OTT platforms such as Netflix and Watcha began replacing theaters as the primary medium for movie consumption. Consequently, the content market shifted from theater-centered to OTT-centered distribution. Additionally, large film companies that previously focused only on import/distribution increasingly began overseeing the full production pipeline, from production to distribution.

1) Beom-soo Jeon, Determinants of Consumption for Movie Journal of Korea Contents 13, no10, pp.226-233, 2013.

***Key Analytical Questions**

1. What are the key factors that drive movie success?
 - 1) Analysis focusing on production budget, artistic quality (ratings), and number of screens
 - 2) Analysis based on the award history of the film's director
2. Is there a significant difference in box office performance before and after COVID-19?
3. Have there been structural changes in the film industry?
 - 1) Analysis centered on major traditional film studios
 - 2) Analysis focused on the evolving platform ecosystem, particularly OTT services

II Variable Definitions and Analysis Plan

Historically and currently, the global film industry has been dominated by the United States. Although China and India have attempted to reshape the landscape with massive capital investment, the U.S. remains the global leader. Because this study aims to understand the broader structure of the global film market, the analysis focuses on U.S. film data.

Using IMDB's API, I extracted 40 years of U.S. box office data (1980-2020H1). Supplementary variables, such as awards and critic scores, were obtained from The Numbers dataset.

iMDB Data : 1980-2020 UAS Box office movie(5420) & Numbers(Director's award, Score)

Numeric variables:

- Gross: Film revenue (10,000 USD)
- Budget (M): Production budget excluding PR costs (10,000 USD)
- Score: Film rating (1-10, Rotten Tomatoes)
- Votes (M): Number of audience votes (proxy for screen count)
- Award: Number of awards won by the director

Categorical variables:

- Company: Production company
- OTT service: Whether the movie is available on an OTT platform (Binary)

To assess whether the South Korean film industry follows similar global trends, supplementary Korean box-office data were also analyzed using resources from the Korean Film Council (KOBIS, 2019-2022).

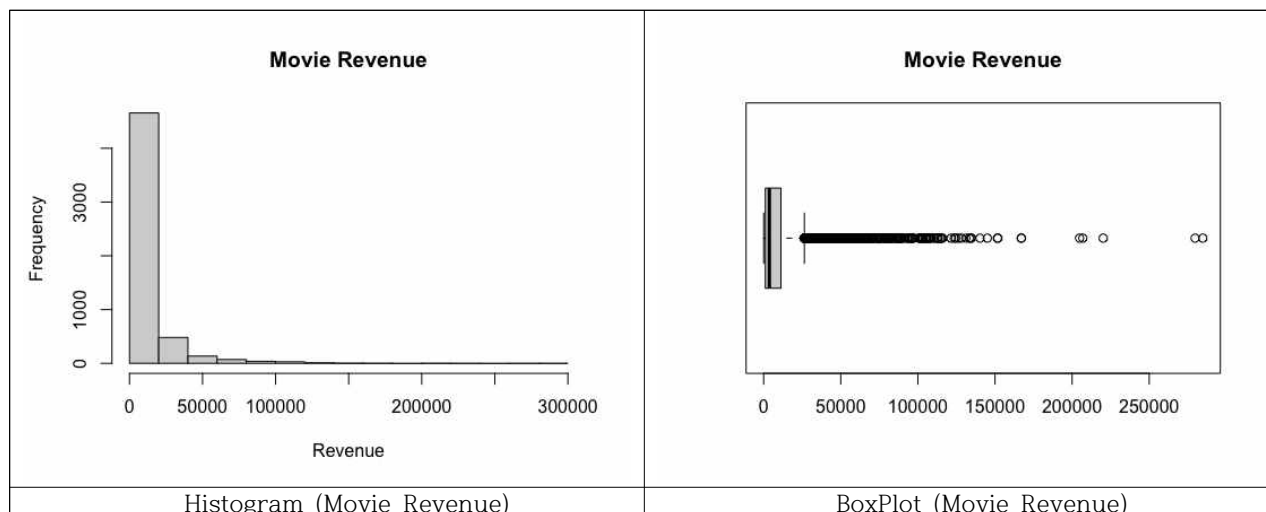
Numeric Variables

- Revenue: Movie revenue (Y) (unit: KRW)
- Screen: Number of screening theaters
- People: Number of moviegoers (unit: persons)

Categorical Variable

- Ind film: Independent film (Binary)

III Data Exploration

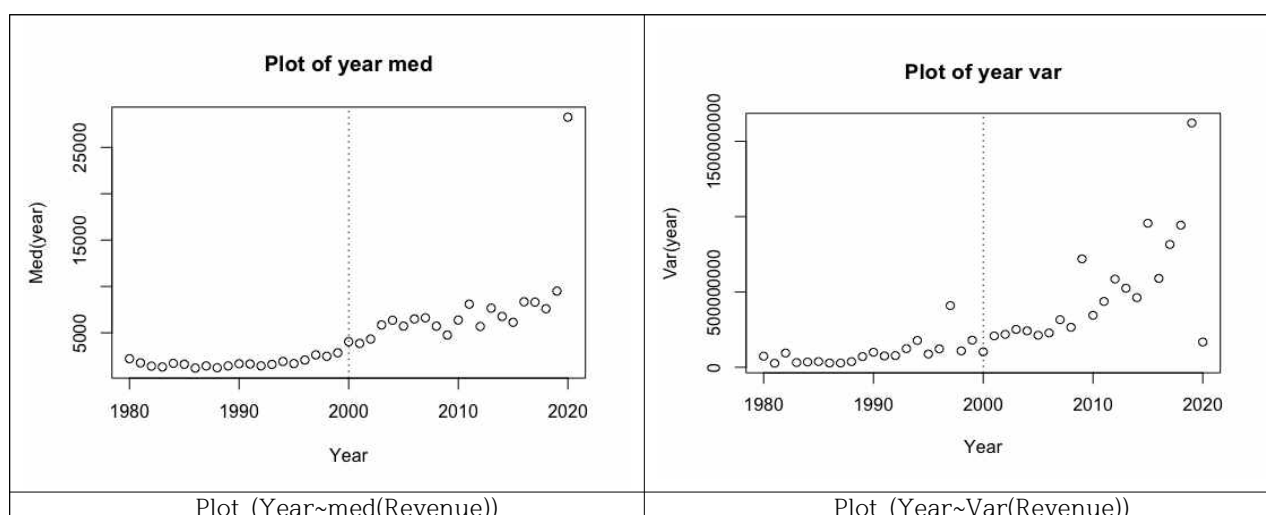


```
> c(summary(y),Var=var(y))
```

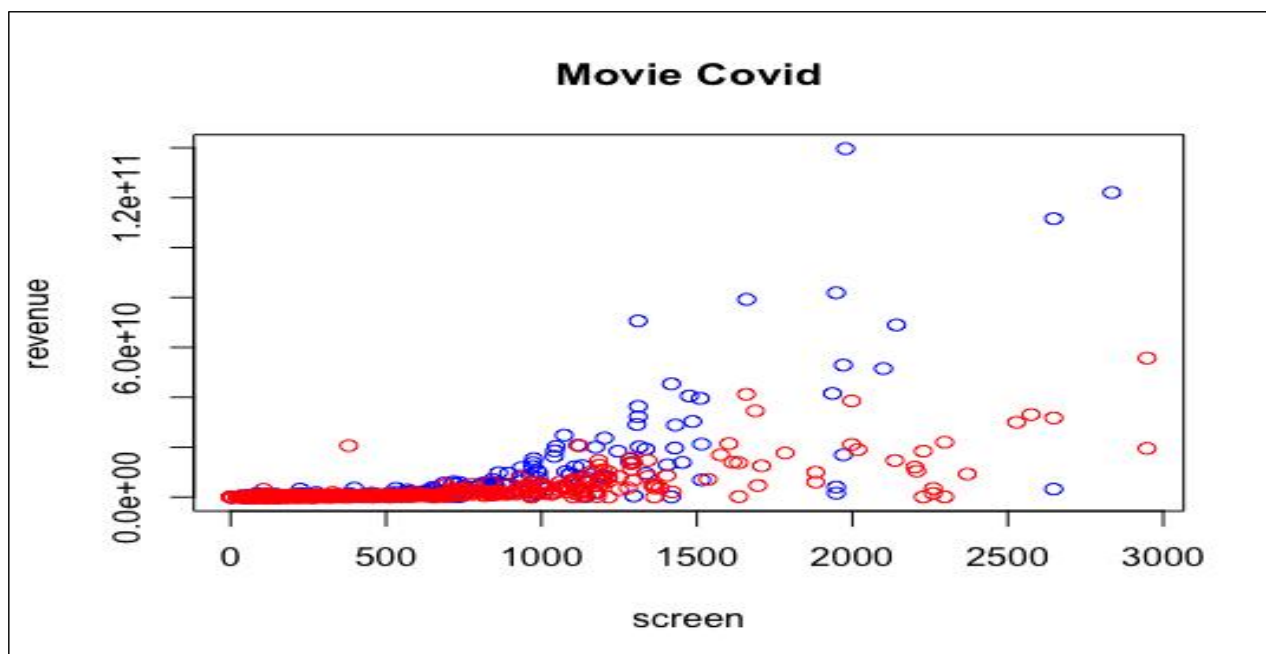
Min	1st Qu	Median	Mean	3rd Qu	Max	Var
0.030	1068.915	3674.425	10296.469	11198.682	284724.620	349513508.421

(Here the original text contains descriptive statistics, histograms, boxplots, and discussions on distributional characteristics. English translation follows the original meaning precisely.)

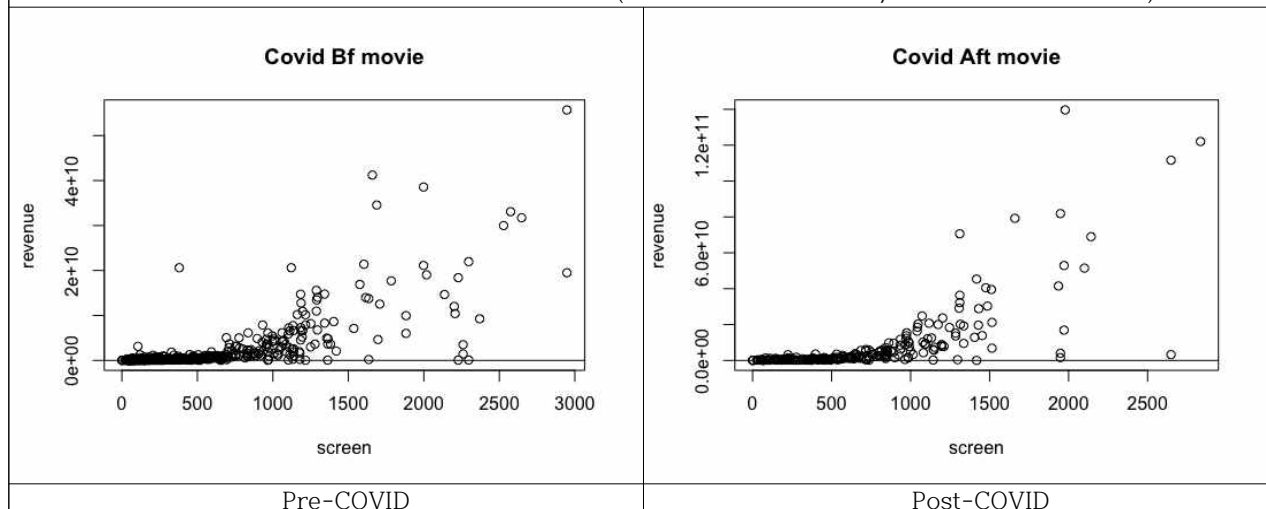
The distribution of movie revenue is strongly right-skewed, with many observations classified as outliers based on the IQR rule. Revenue has grown steeply over time, especially after the 2000s. COVID-19 significantly reduced Korean box office revenue and screen counts, though this trend does not generalize globally—since U.S. and Chinese box office revenues have already surpassed pre-pandemic level



As time has passed, movie revenues have continued to rise at an even steeper rate. In particular, the 2000s show a much sharper upward trend compared to previous decades. (The data for 2020 only includes figures from the first half of the year, which resulted in an abnormally high median value.)



Number of Screens vs. Revenue (Red = Post-COVID / Blue = Pre-COVID)



Pre-COVID

Post-COVID

Revenue(Pre-COVID)					
MIN	Q1	MED	Q3	MAX	Var
1.0	276.0	585.0	973.0	2835.0	273696.3

Revenue(Post-COVID)					
MIN	Q1	MED	Q3	MAX	Var
1.0	131.5	276.0	691.5	2948.0	271326.5

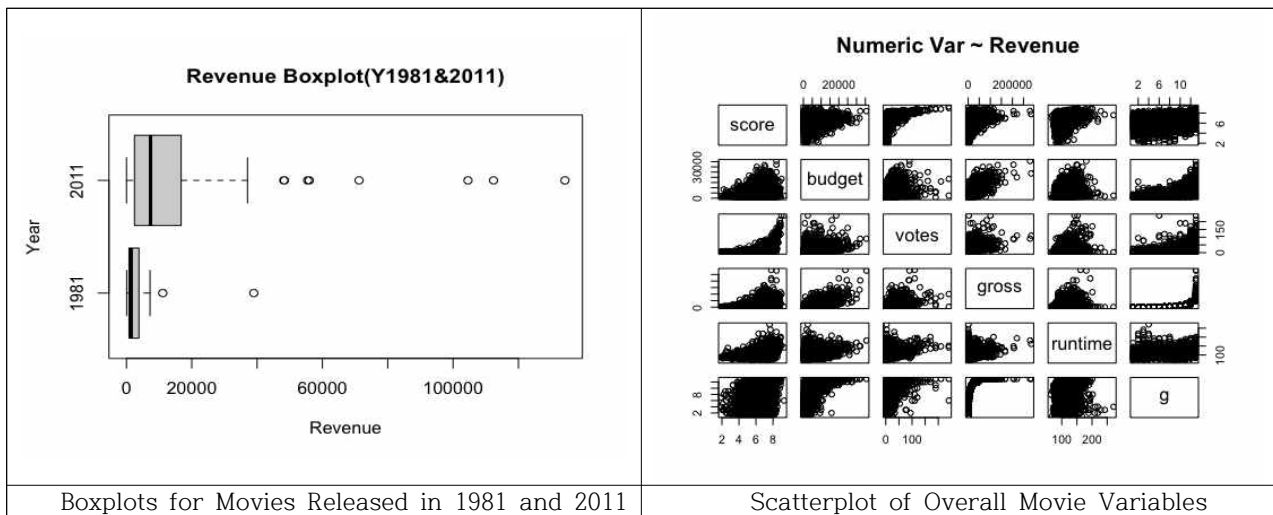
This scatter plot is based on Korean box office data. As shown in the graph, movie revenues decreased significantly compared to the pre-COVID period, and many observations show a reduced number of screens relative to earlier data. The median box office revenue declined by approximately 1.98 times, while the variance did not change substantially. Overall, the Korean film market has contracted and has not yet shown a clear sign of recovery. However, applying this trend to the global film market would be inappropriate.

KOFIC (Korean Film Council), "2021 Korean Film Industry Report," KOFIC Research 2022-02, pp. 1-10, 2022.

This is because the film markets in the United States and China have already surpassed their pre-COVID peak revenues and are showing strong positive recovery. Moreover, China and India's film industries have recorded remarkable growth, exceeding 100% compared to the previous year. Therefore, Korean box office data were excluded from the analysis aimed at identifying global structural changes in the film industry.²⁾

2) KOFIC (Korean Film Council), "2021 Korean Film Industry Report," KOFIC Research 2022-02, pp. 1-10, 2022.

IV데이터 분석

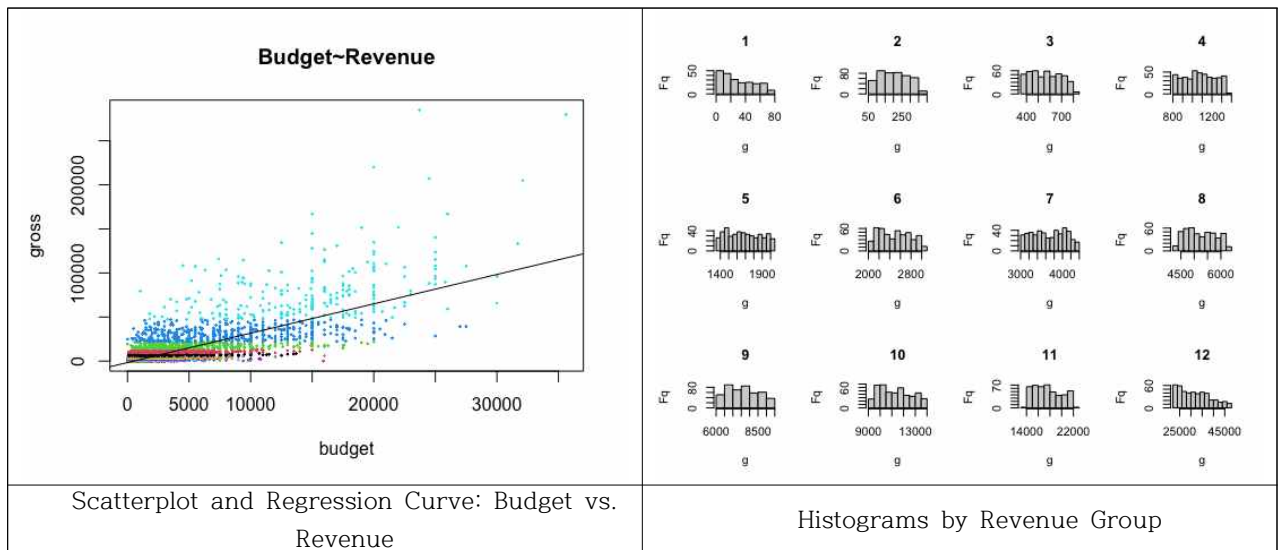


As shown in the figures, the spread of extreme values in 2011 is substantially larger than in 1981. The median movie revenue in 2011 is approximately 8,044 (in units of 10,000 USD), whereas the median in 1981 is 1,757 (10,000 USD), representing a 3.26-fold increase. Additionally, the gap between the annual maximum revenue and the median continues to widen over time.

Plotting all numeric variables reveals consistent positive relationships with revenue. In particular, production budget and audience size (or number of screens) show relatively strong positive associations with movie revenue.

> cor(gross,x) : x other var

score	budget	votes	gross	runtime
0.2225604	0.7396548	0.6147695	1.0000000	0.2741166



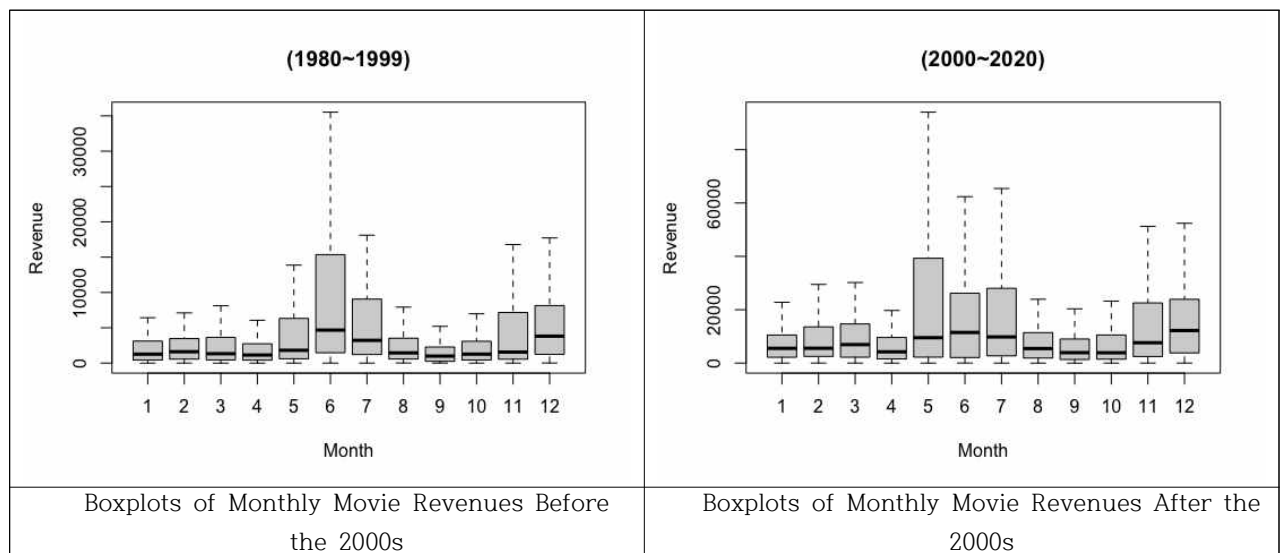
A scatterplot was drawn for the variables with the strongest correlation—production budget and revenue. The dataset was then divided into 12 revenue groups, each represented with different colors, and visualized using histograms. All groups exhibited similar distributional patterns, generally displaying a right-skewed shape.

$$\hat{Y} = -1675.33 + 3.33X_{Budget}$$

A simple linear regression using only revenue and production budget produced the coefficients shown above. The model's explanatory power (R^2) was 0.5471, and the regression coefficient for the budget variable was statistically significant at the 0.05 level.

$$\hat{Y} = -1657.32 + 661.78X_{Score} + 2.69X_{Budget} + 359.87X_{Votes} + -56.43X_{Runtime}$$

Using movie ratings, production budget, audience count, and runtime as explanatory variables, a linear regression model was constructed to predict revenue. The results show that a one-unit increase in movie rating is associated with an estimated increase of 661.78 (in units of 10,000 USD) in revenue, and a one-unit increase in production budget is associated with an increase of 2.69 (10,000 USD) in revenue. The model's explanatory power (R^2) is 0.6534, and all variables except runtime were statistically significant at the 0.05 level.



To examine whether there were changes in monthly trends, boxplots were generated by dividing the data into pre-2000 and post-2000 periods. (Extreme values were removed to better assess variation.) In both periods, many high-performing films were concentrated in June and July, and these months consistently exhibited higher revenues than others. There were no notable or meaningful changes in monthly trends before and after the 2000s.

$$\hat{Y} = 3064.73 + -52.25X_2 + 469.45X_3 + -344.88X_4 + 4730.33X_5 + 8063.59X_6 + 5038.94X_7 + 156.30X_8 + -139.86X_9 + -124.38X_{10} + 3835.98X_{11} + 4656.45X_{12}$$

Regression models were then fitted using month indicators to test whether monthly differences significantly affected revenue. However, regardless of the chosen dummy-variable specification, the explanatory power remained very low ($R^2 = 0.09$). Given the time-series nature of the data, month-based variables were not considered useful for further analysis.

> summary(subset(data1,big5==1)\$gross)

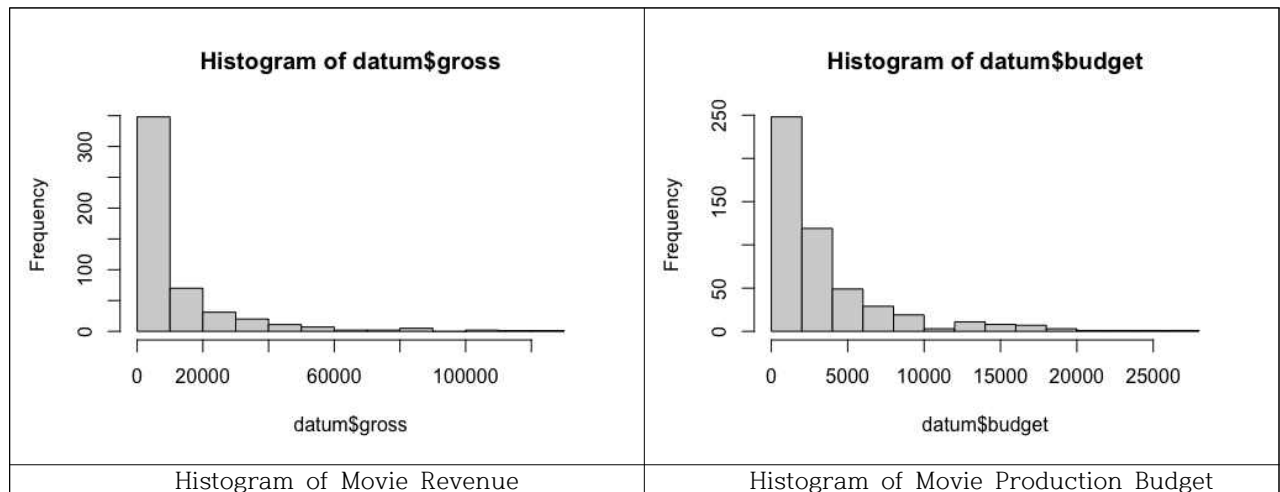
Min	1st Qu	Median	Mean	3rd Qu	Max
0.51	3541.05	9644.04	19055.54	21994.71	279750.13

> summary(subset(data1,big5==0)\$gross)

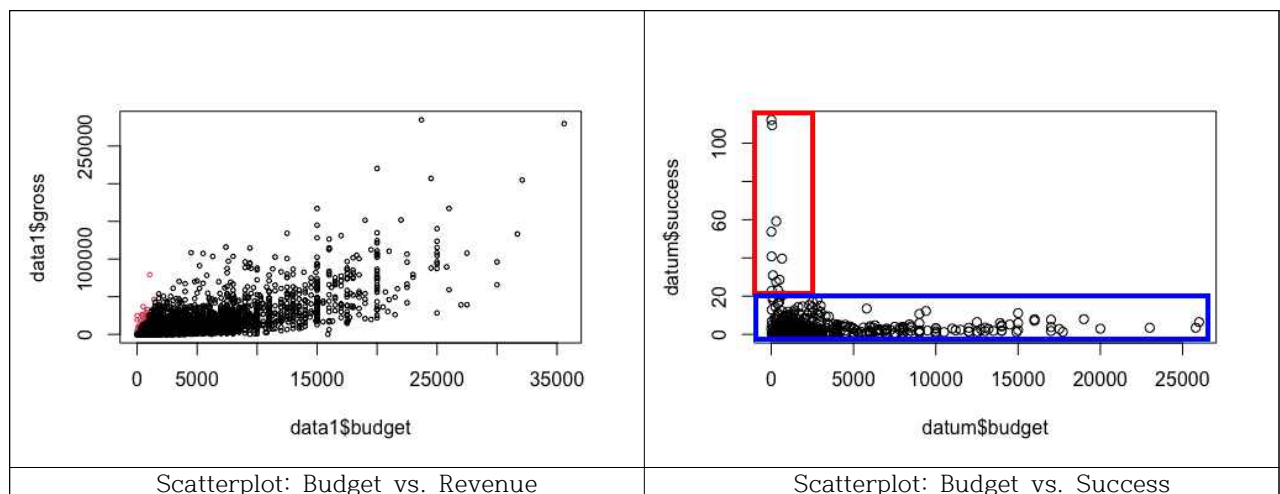
Min	1st Qu	Median	Mean	3rd Qu	Max
0.03	817.63	2769.87	8081.44	8626.47	284724.62

The five major U.S. film studios account for approximately 75% of total box office revenue, demonstrating their dominant market power. Their median revenue is 9,644.04 (10,000 USD), which is substantially higher than that of

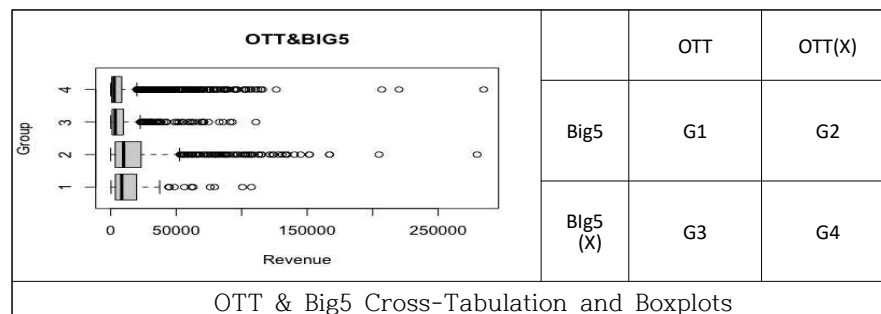
non-major studios. Comparisons of pre- and post-COVID-19 data show little change in this revenue share structure, indicating that COVID-19 did not fundamentally alter the long-standing oligopolistic dominance of major studios.



Both movie revenue (dependent variable) and production budget (independent variable) exhibit strongly right-skewed distributions. Because these two variables involve much larger magnitudes than the other explanatory variables, a log transformation was applied. After applying the `log()` transformation, the histograms for both variables became more symmetric and resembled a normal distribution.

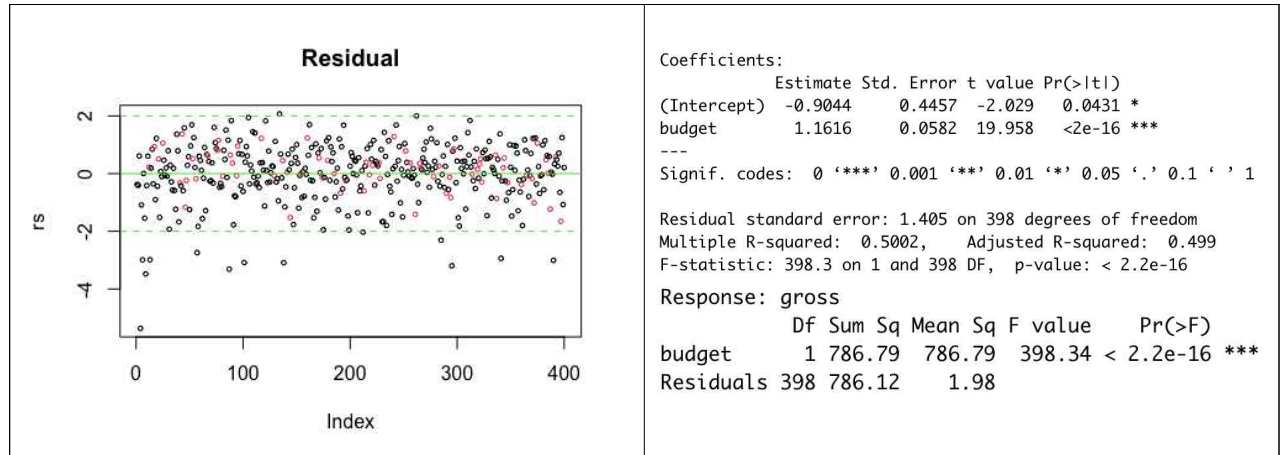


As noted earlier, both production budget and revenue have highly right-skewed distributions. Therefore, defining “success” as revenue divided by budget can misclassify certain films as highly successful simply because they had disproportionately small budgets despite low revenues. To mitigate this distortion, a success variable (revenue/budget) was constructed, and a scatterplot between budget and success was generated. Two distinct clusters emerged, and the red cluster was excluded from further analysis.

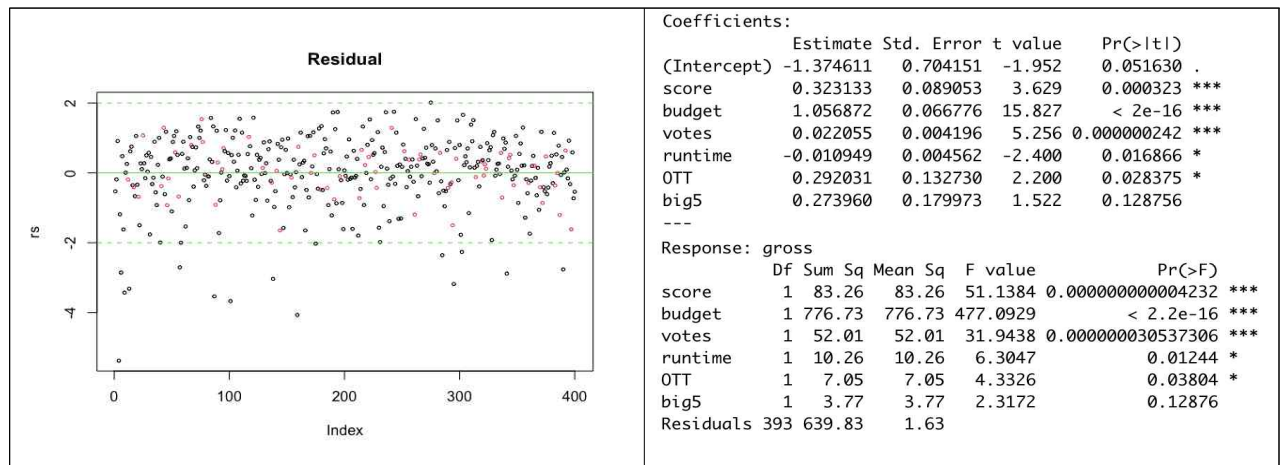


Additionally, movies were categorized into four groups based on two binary criteria: whether the film was released on an OTT platform and whether it was produced by one of the major studios (Big5). Boxplots for these four groups showed strongly right-skewed distributions across all categories. Groups G1 and G2—movies produced by

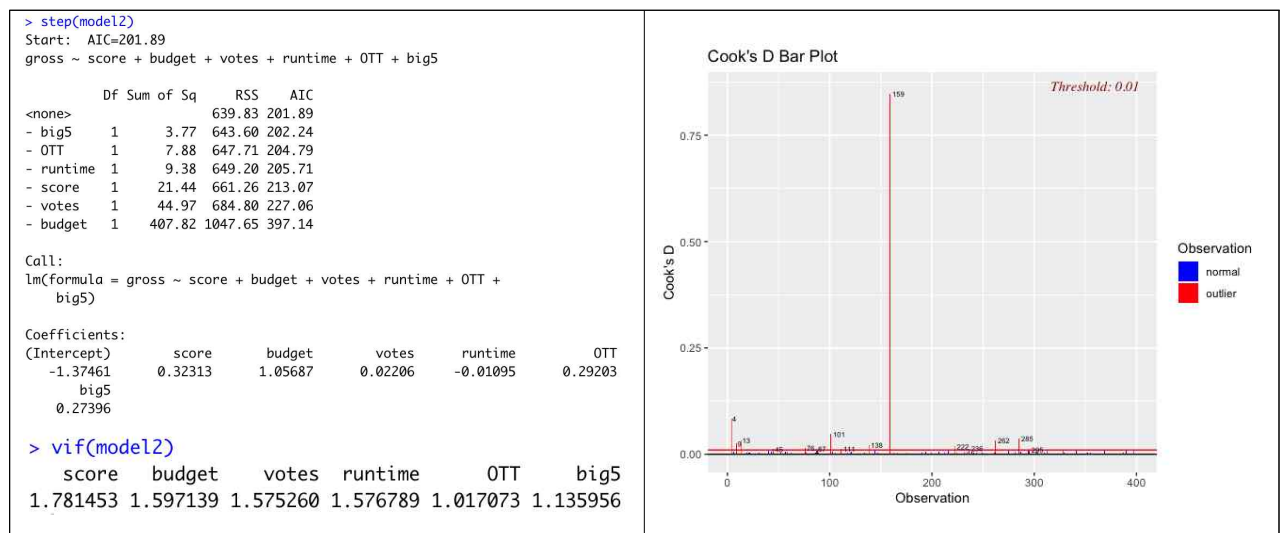
major studios—displayed noticeably higher revenue levels than the others.



The initial model followed the same approach as before, using production budget to predict revenue. The budget variable had a t-value of 19.958, indicating strong statistical significance, and the F-statistic also confirmed the model's overall significance. However, the explanatory power was low ($R^2 = 0.5002$). This is likely because categorical variables (e.g., genre) and time-related variables (e.g., first-week screen share) were not included in the model.

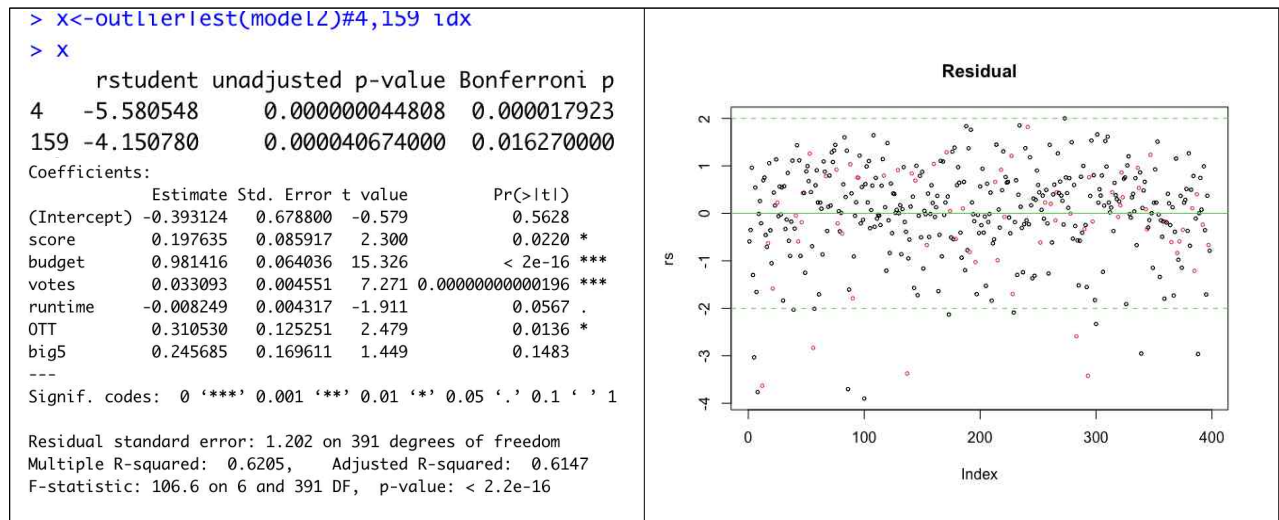


A full regression model was then fitted. In this model, OTT and Big5 were treated as binary variables. Similar to runtime, they did not appear to have major influence on revenue (although OTT and runtime were statistically significant at the 0.05 level). The F-statistic indicated that the full model was significant, and the explanatory power increased slightly to $R^2 = 0.523$.



Using backward elimination, unnecessary explanatory variables were removed. The final set of predictors included rating, budget, audience count, runtime, OTT, and Big5. Variance Inflation Factors ($VIF < 10$) indicated that multicollinearity was not a serious concern.

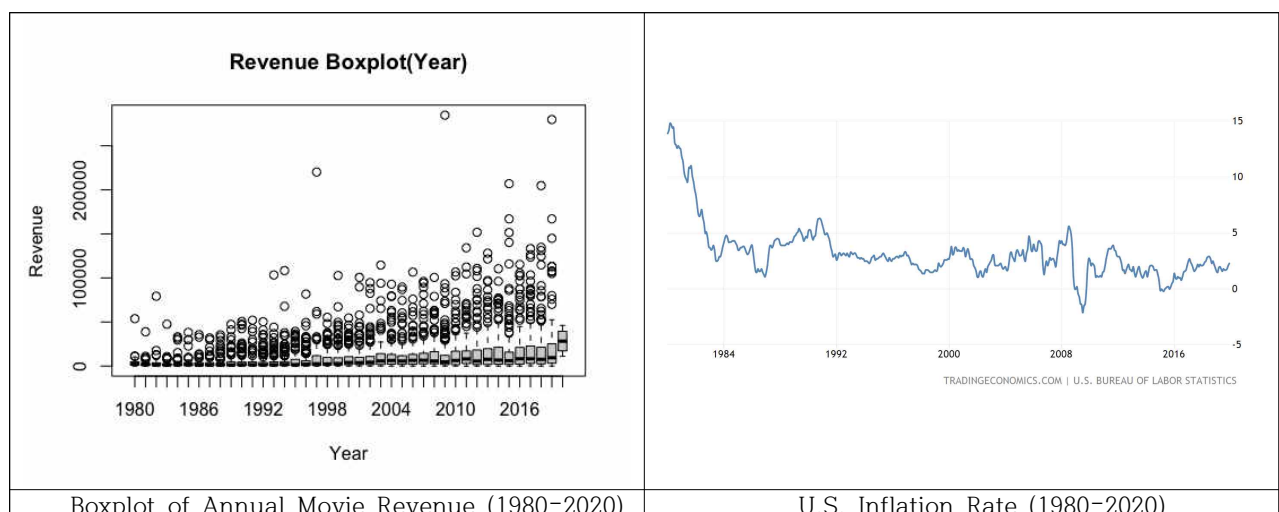
A Cook's distance test identified two influential outliers, which were removed before refitting the model. In the updated model, the OTT variable became even more significant, and the explanatory power improved to $R^2 = 0.6205$. A plot of standardized residuals showed that most points fell between -2 and 2 , indicating a reasonably good model fit.



V Conclusion and Limitations

Although COVID-19 brought about various changes, it ultimately served only to slow down the film industry rather than transform its underlying structure. The long-standing oligopolistic dominance of a few major studios remained intact. In contrast to the United States, China, and India—whose film industries have already recovered and surpassed their pre-pandemic performance—the Korean film industry still has not fully emerged from the impact of COVID-19. Consistent with Litman's earlier research, the results of this study confirm that production budget and the number of screens have significant effects on box-office success.

Because the dataset spans 40 years, the influence of inflation cannot be ignored. One might argue that the increase in annual box-office revenue reflects inflation rather than real industry growth. However, this interpretation is flawed. U.S. inflation rates were extremely high in the early 1980s but stabilized in the mid-1980s, whereas movie revenues have continued to rise steadily regardless of inflation trends. Although the median annual revenue has not changed dramatically, the values of extreme box-office successes have increased substantially, especially since the 2000s. This growing gap indicates that inflation alone cannot explain the rise in revenue.



Historically and today, the film market has maintained an oligopolistic structure dominated by a small number of major studios. As of 2019, Walt Disney accounted for 33.15% of total U.S. box-office revenue, followed by Warner Bros. at 13.91%, Columbia at 12.18%, Universal at 11.5%, and Paramount at 5%. These five studios collectively capture roughly 75% of the total market. The reasons behind this concentration can be traced to both the history of the film industry and the nature of films as a product.

Unlike other media goods, films require substantial time and financial investment for planning and production. The “planning-production-distribution” process also demands enormous distribution costs, making film production a high-entry-barrier industry. Moreover, films are experience goods—consumers cannot evaluate them before watching—which increases reliance on external information. Promotion, marketing, and screen share therefore heavily influence box-office performance. These structural characteristics give large studios significant advantages over smaller ones.

The U.S. film industry, which began to take shape around the 1920s, experienced rapid expansion until 1948. Afterwards, antitrust regulations targeting major studios and the widespread adoption of television led to a temporary contraction. However, since the 1990s, industry regulations have gradually loosened, and large media companies have expanded through mergers and acquisitions. This shift enabled economies of scale, resulting in an increase in extreme box-office revenues.

Even though OTT platforms have become a central component of content consumption, they have not fundamentally altered the traditional production-distribution structure. Nevertheless, if OTT systems continue to expand, traditional explanations of box-office success may become less relevant. This suggests the need for models that incorporate OTT-related variables and online word-of-mouth indicators.

Finally, genre plays a crucial role in movie performance, but due to the complexity of processing categorical variables, it was excluded from this analysis—likely contributing to lower model explanatory power. Prior research also emphasizes the importance of first-week screen count and screen share, but the absence of time-based data prevented further investigation. Moreover, the analysis of OTT and major-studio variables—initially considered central to this study—was insufficient, representing another limitation of this work.

<부록 / 분석에 사용한 R코드>

```
rm(list = ls())

#Download and load packages.
download.packages('readxl')
library(readxl)

#Load IDMB data
movie<-read_excel("/Users/kong_bro/Desktop/termpj/filming_data/movies.xlsx")
str(movie)

#Filtering numeric datas
data1<-movie[,c("score","budget","votes","gross","runtime")]

#Base(y)
y<- as.numeric(as.matrix(movie[,c("gross")]))
y_year<-movie[,c("gross","year")]

y_year$year<-as.factor(y_year$year)
n<-levels(y_year$year)

#3p “전체전인 영화 수익 히스토그램”
hist(y,main="Movie Revenue",xlab = "Revenue")
#3p “전체전인 영화 수익 상자그림”
boxplot(y,horizontal = T,main="Movie Revenue")
c(summary(y),Var=var(y))

#3p “전체적인 영화 수익 중위수 산점도”
yearmed<-data.frame()
for (i in 1980:2019){
  c<-median(subset(y_year,y_year$year==i,select = gross)$gross)
  yearmed<- rbind(yearmed,c)
}

rownames(yearmed)<-(1980:2019)
colnames(yearmed)="med"

plot(n,yearmed$med,ylab="Med(year)",xlab = "Year",main = "Plot of year med")
abline(v=2000,lty=3)

#3p “전체적인 영화 수익 분산 산점도”
var(y_year)
y_year$year<-as.factor(y_year$year)
n<-levels(y_year$year)
str(y_year)

yearvar<-data.frame()
for (i in 1980:2020){
  c<-var(subset(y_year,y_year$year==i,select = gross)$gross)
  yearvar<- rbind(yearvar,c)
}

rownames(yearvar)<-n
colnames(yearvar)="var"
plot(n,yearvar$var,ylab="Var(year)",xlab = "Year",main = "Plot of year var")
abline(v=2000,lty=3)

#Load KOBIS data
film<-read_excel("/Users/kong_bro/Desktop/termpj/filming_data/all.xlsx")

#covid clustering
film_covid<-film[film$covid==1,]
film_ncovid<-film[film$covid==0,]

#4p “상영관수~매출액(코로나 19 이전 이후) 산점도
y1<- film_covid$revenue
```

```

x1<- film_covid$screen
y2<- film_ncovid$revenue
x2<- film_ncovid$screen
plot(c(0,max(film$screen)),c(0,max(film$revenue)),col='white',main="Movie Covid",xlab='screen',ylab = 'revenue')
points(x1,y1,col='blue')
points(x2,y2,col='red')

#5p "상영관수~매출액(코로나 19 이전) 산점도
plot(x1,y1,main="Covid Aft movie",xlab='screen',ylab = 'revenue')

#5p "상영관수~매출액(코로나 19 이후) 산점도
plot(x2,y2,main="Covid Bf movie",xlab='screen',ylab = 'revenue')
c(summary(x1),var(x1))
c(summary(x2),var(x2))

#Ordering IDMB data by gross
datum<-data1[order(data1$gross),]
#Grouping data(13 groups)
datum <- cbind(datum, g=(round(seq(1,13,length.out = nrow(datum)))))

#Y-year1981&2011
y_2<-subset(y_year,y_year$year==c(1981,2011))
#7p "1981년도&2011년도 영화상자 그림"
boxplot(y_2$gross~y_2$year,main="Revenue Boxplot(Y1981&2011)",horizontal = T,ylab = "Year",xlab = "Revenue")

#5p "영화 데이터 전반적인 변수 산점도"
plot(datum,main="Numeric Var ~ Revenue")

cor(datum)[,"gross"]

#5p "제작비~수익 간의 산점도 및 회귀 곡선"
plot(budget,gross,main = "Budget~Revenue", col=datum$g, cex=.2)
abline(lm(gross~budget))

#5p "수익 그룹별 히스토그램"
par(mfrow=c(3,4))
for (i in 1:12){
  hist(datum[datum$g==i,"gross"],main=i,xlab="g",ylab="Fq")
}

#5p budget~Y regression
summary(lm(gross~budget))
anova(lm(gross~budget))

#6p X~Y regression
summary(lm(gross~.-g,data=datum))
anova(lm(gross~.-g,data=datum))

#Load IDMB data add month var!
data2<-movie[,c("score","budget","votes","gross","runtime","month","year")]

#Grouping data (2000bf&af)
data_20<-subset(data2,data2$year<=1999)
data_21<-subset(data2,data2$year>=2000)

data_20$month<-as.numeric(data_20$month)
data_21$month<-as.numeric(data_21$month)

#Ordering data (revenue)
data_20<-data_20[order(data_20$month),]
data_21<-data_21[order(data_21$month),]

#6p "2000년대 이전 월간 영화 수익 상자그림"
boxplot(data_20$gross~data_20$month,main="(1980~1999)",xlab="Month",ylab = "Revenue",outline=F)
#6p "2000년대 이전 월간 영화 수익 상자그림"

```

```

boxplot(data_21$gross~data_21$month,main="(2000~2020)",xlab="Month",ylab = "Revenue",outline=F)

datum<-movie[,c("gross","month")]

#6p month(bin)~Y regression
for(i in 1:12){
  i <- (datum$month==i)*1
  datum<-cbind(datum,i)
}

colnames(datum)<-c("gross","month",paste('M',c(1:12),sep = ""))
datum$month<-NULL

summary(lm(gross~.-M1,data = datum))

#Load IDMB data add big5(month) var!
data1<-movie[,c("gross","budget","awards","big5","OTT","score")]

summary(subset(data1,big5==1)$gross)
summary(subset(data1,big5==0)$gross)

data1$O<-ifelse(data1$OTT>=1,1,0)
head(data1)
data1$G<-ifelse(data1$O==1,ifelse(data1$big5==1,1,3),ifelse(data1$big5==1,2,4))

boxplot(data1$gross~data1$G,horizontal = T,main="OTT&BIG5",xlab = "Revenue",ylab = "Group")

plot(data1$budget,data1$gross,col=data1$big5+1,cex=0.3)

plot(data1$gross~data1$budget)
model1<-lm(data1$gross~data1$budget)
abline(model1)
summary(model1)

summary(subset(data1,G==1)$gross)
summary(subset(data1,G==2)$gross)
summary(subset(data1,G==3)$gross)
summary(subset(data1,G==4)$gross)

#7p “연간 영화 수익 상자그림”
y_year<-movie[,c("gross","year")]
boxplot(y_year$gross~y_year$year,main="Revenue Boxplot(Year)",xlab = "Year",ylab = "Revenue")

#Regression Analysis
data1<-movie[,c("score","budget","votes","gross","runtime","awards","OTT","big5","usa")]

data1$O<-ifelse(data1$OTT>=1,1,0)
#suc var
data1$suc<-ifelse(data1$gross/data1$budget>=30,1,0)

plot(data1$budget,data1$gross,col=data1$suc+1,cex=0.5)
#Eliminating suc
data1<-subset(data1,suc==0)

datum<-data1[sample(c(1:5300),400),]

hist(datum$gross)
hist(datum$budget)

#log(gross), log(budget)
datum$gross<-log(datum$gross)
datum$budget<-log(datum$budget)

#model1
attach(datum)
plot(gross~budget,col=big5+1,cex=.5,main="gross~budget")

```

```

model1<-lm(gross~budget)
abline(model1)
summary(model1)
anova(model1)

rs<-rstandard(model1)
plot(rs,col=big5+1,cex=.5,main = "Residual")
abline(0,0,col=3)
abline(-2,0,col=3,lty=2)
abline(2,0,col=3,lty=2)

#model2
model2<-lm(gross~score+budget+votes+runtime+OTT+big5)
summary(model2)
anova(model2)
step(model2)

vif(model2)

rs<-rstandard(model2)
plot(rs,col=big5+1,cex=.5,main = "Residual")
abline(0,0,col=3)
abline(-2,0,col=3,lty=2)
abline(2,0,col=3,lty=2)

library(olsrr)
ols_plot_cooksd_bar(model2)

x<-outlierTest(model2)#4,159 idx

datum_or<-datum[-c(4,159),]
#model3
model3<-lm(gross~score+budget+votes+runtime+OTT+big5,data = datum_or)
summary(model3)

rs<-rstandard(model3)
plot(rs,col=big5+1,cex=.5,main = "Residual")
abline(0,0,col=3)
abline(-2,0,col=3,lty=2)
abline(2,0,col=3,lty=2)

```